# UM-2 solution

An accurate approximate solution of the coupled field equations, with Kohn-Sham theory for the dirac16complex field coupled to gravity and to the Standard-Model fields, three generations of quarks and leptons, and the flavour conversion of the three neutrinos

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Solution, version 2, 2026-10-18 (version 1: 2026-10-17)

Repository once-ere/Dirac_claude, folder universe_model_2

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# 1 How to read this document

This document presents the approximate solution of the coupled field equations of UM-2 and the verification of each step. It is self-contained: every equation it uses is written here. Labels: AUTHOR, RECORD, STANDARD, ASSUMED, INVENTED, DERIVED (computed and verified; the verification is stated), ESTIMATE, OPEN. Units $\hbar = c = 1$; in the numerical work $H = 1$ and $m = H$ unless a physical unit is given. The coordinates are the author's $x_0, x_1, \ldots, x_7$ ($x_0$ the hidden space direction, $x_1, x_2, x_3$ space, $x_4$ the time, $x_5, x_6, x_7$ the extra times); $W^+$ carries the electric charge $+e$ and $W^-$ the charge $-e$.

Version 2 (2026-10-18) has three generations of quarks and leptons and three neutrino flavours with their mixing; step 5 is the solution for the flavour conversion of the neutrinos (vacuum, matter, the Sun, accelerators, the early Universe and $N_{eff}$, the cosmic neutrino background, supernovae). The chiral tip $Z$ portal of version 1 is removed (step 4).

Symbols with more than one use: $k$ is the block variable $1/f^2$ ($k_*$, $\delta k$, $G(k)$), the Lovelock order as the index of $\alpha_k$, $E_{(k)}$, $\mathcal{L}_{(k)}$ (written $p$ where the block variable appears in the same formula), the momentum of a mode along $x_1, x_2, x_3$ in $h_j$, $\vec{k}$, $\Delta k$ and $\varepsilon = e^{-a_4}|k|$, the comoving Fourier wavenumber in the perturbation equations of step 6, and the wave number of a flavour mode in step 5. $\Lambda = -10H^2$ is the eight-dimensional cosmological constant and $\Lambda_4$ the four-dimensional one; in the $x_0$ eigenvalue problems of step 2, $\Lambda = (m\,fL)^2$ is a dimensionless eigenvalue; $\Lambda_\star$ is the scale of the effective interactions. $n$ is a number density, except as a level number ($n\pi$) and in the lattice vector $\vec{n} \in \mathbb{Z}^3$ of a momentum shell; $N$ is a particle number. The subscript $t$ ($p_t$, $v_t$) refers to the extra times $x_5, x_6, x_7$; $\xi = y/L$ is the scaled hidden coordinate; primes denote $d/d\xi$ in step 2 and $d/d\tau$ (conformal time) in step 6. $f$ is the block scale and $f_o$ an occupation number. $\rho$ is an energy density, except for the electroweak parameter $\rho = \Lambda_W/(\Lambda_Z \cos^2\theta_W)$ (measured value $\rho_0$), and $\theta_W$ is the angle of the gauge couplings. $w$ denotes the Lovelock weights ($w_2$, $w_3$), the weight 1/2 of a fold parity in the Kohn-Sham sums, $\sqrt{c^2 - e^2}$ in the fermion mass condition and an equation-of-state parameter. $\psi$ is a Standard-Model fermion field, in step 5 a flavour amplitude and in step 6 the Newtonian potential; $\sigma$ is a Pauli matrix ($\sigma_1, \sigma_2, \sigma_3$), the annihilation cross-section ($\sigma v$), $\sigma_{SI}$ the scattering cross-section, $\sigma_b$ the cosmic variance of a bin of multipoles and, in step 6, the shear; $t_o$ is an orbital density. Flavour: $i = 1, 2, 3$ is the generation index and labels the neutrino mass states, $\alpha \in \{e, \mu, \tau\}$ a flavour index (a colour index on $G^\alpha$); $U$ is the PMNS and $V$ the CKM matrix (the matter potentials carry subscripts, $V_{CC}$, $V_{NC}$, $V_{\nu\nu}$; $V_\perp$ is a volume); $\kappa_\nu$, $\kappa_e$, $\kappa_u$, $\kappa_d$ are the tip matrices; $\mu = \sqrt{2} G_F n_{\nu_e}$ (no subscript) is the unit of the neutrino self-interaction, $\mu_\psi$ a bulk mass and $\mu$ the muon; $e$ is the elementary charge, the electron and the eigenvalue $\varepsilon f L$; $J$ the Jarlskog invariant (and the block operator $\gamma^{x_0}\gamma^{x_1}\gamma^{x_4}$); $s_{ij} = \sin\theta_{ij}$, $c_{ij} = \cos\theta_{ij}$ ($c$ with the subscripts $L$, $Q$, $R$, $\psi$ a localisation); $u = \cos\vartheta$ the direction cosine of a neutrino in step 5 ($u$ also the up quark).

# 2 The equations being solved

The action of UM-2 is

$$S = \frac{1}{2\kappa}\int d^8x\sqrt{|g|}\Big(\sum_{k=1}^{3}\alpha_k\mathcal{L}_{(k)} - 2\Lambda\Big) - \int d^8x\sqrt{|g|}\,\rho_b(f) + S_\Psi + S_{gauge} + S_f + S_{tip} + S_{I11} + S_{I12},$$

with the author's $\alpha_k = w_k H^{2-2k}$, $w = (1, \frac{1}{32}, \frac{1}{1152})$, $\Lambda = -10H^2$; the stabiliser $\kappa\rho_b = \kappa\rho_b^*(f_*/f)^8$; the dirac16complex Lagrangian $\sqrt{|g|}[\frac{1}{2}(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi) - m\bar\Psi\Psi - \frac{\lambda}{2}(\bar\Psi\Psi)^2]$ with the canonical spin connection; the eight-dimensional $SU(3)_c \times SU(2)_L \times U(1)_Y$ gauge fields and three generations of fermions, $\psi_i \in \{Q_i, u_{Ri}, d_{Ri}, L_i, e_{Ri}, \nu_{Ri}\}$ ($i = 1, 2, 3$), with bulk masses $\mu_{\psi i}$ (the doublets of a sector sharing one, $\mu_{Li} = \mu_L$, $\mu_{Qi} = \mu_Q$); the tip sigma field $\Sigma \in SU(2)$ (no Higgs particle) with the tip Yukawa couplings $-\sum_{ij}(\bar L_i\Sigma(\Lambda_\ell)_{ij}\ell_{Rj} + \bar Q_i\Sigma(\Lambda_q)_{ij}q_{Rj} + \text{h.c.})$ of complex $3\times3$ generation matrices; and the portals

$$S_{I11} = \int d^8x\sqrt{|g|}\,\frac{c_V}{\Lambda_\star^6}(i\bar\Psi\gamma^M\Psi)\,\mathcal{J}_M, \qquad \mathcal{J}_M = \sum_{i=1}^{3}\sum_{\psi} q_\psi(i\bar\psi_i\gamma_M\psi_i),$$

$$S_{I12} = -\int d^8x\sqrt{|g|}\,\frac{\bar\Psi\Psi}{\Lambda_\star^3}\,\mathcal{F}, \qquad \mathcal{F} = c_G G^\alpha_{MN}G^{\alpha MN} + c_W W^a_{MN}W^{aMN} + c_B B_{MN}B^{MN}.$$

The field equations are

$$\sum_k \alpha_k E_{(k)}{}^M{}_N + \Lambda\delta^M_N = \kappa T^M{}_N, \qquad \gamma^M D_M\Psi = \Big[m + \lambda\bar\Psi\Psi + \frac{\mathcal{F}}{\Lambda_\star^3}\Big]\Psi - i\frac{c_V}{\Lambda_\star^6}\mathcal{J}_M\gamma^M\Psi,$$

together with the Yang-Mills equations with fermion, tip and portal currents, the Dirac equations of the fermions with the tip Yukawa terms, and the boundary conditions at the fold and the tip.

# 3 The approximation scheme

The solution uses the small parameters of the problem:

- $\epsilon \sim 10^{-5}$: the metric is $g = \bar g + \epsilon h^{(1)} + \dots$ (order 0: the background; order 1: the CMB, CNB, BAO);
- $(A/H)^2 \ll 1$ after the primordial era: the matter sources are tiny compared with the stabiliser (today $10^{-92}$), so the block responds adiabatically and linearly;
- the Standard-Model masses are far below the gap $3H/(2f)$ of the $x_0$ operators: four-dimensional effective fields;
- dirac16complex is treated by Kohn-Sham density-functional theory on the static block, with the portal couplings as external mean fields (the Standard-Model fields enter through their densities and condensates).

The hierarchy: (1) the background and the primordial era; (2) the $x_0$ spectra, i.e. all masses and the mixing matrices; (3) the Kohn-Sham states of $\Psi$ coupled to gravity and to the Standard-Model fields, and their back-reaction on the block; (4) the electroweak sector; (5) the flavour conversion of the three neutrinos, $N_{eff}$ and the CNB; (6) the linear cosmology; (7) the dark-matter relic.

# 4 Step 1: the background

## 4.1 The static block

The order-0 metric is

$$ds^2 = e^{2a_4}d\vec x^{\,2} - dx_4^2 + f^2\big[dy^2 - e^{2Hy}d\vec x_t^{\,2}\big], \qquad y = \frac{\ln|\sin 6Hx_0|}{6H}, \quad f = e^{-a_4}b_4^{1/2},$$

periodic in $x_0$ with period $\pi/(6H)$ (two halves joined at the fold $y=0$, a regular tip at $y=-L$, $\eta = HL = 3$). With $k = 1/f^2$, $A = \dot a_4$, $h = \dot f/f$, a static block ($h=0$, $A=0$) with the stabiliser ($p_3 = -\rho_b$, $p_0 = p_t = \rho_b$) requires (DERIVED)

$$\kappa\rho_b = 10H^2 - 6H^2k + \tfrac{3}{8}H^2k^2 = 3H^2k - 10H^2 \implies k_* = 12 + \tfrac{4}{3}\sqrt{51} = 21.5219,$$

$$f_* = 0.215556, \qquad G(k_*) = 1 - 24w_2k_* + 72w_3k_*^2 = \tfrac{20}{3} + \sqrt{51} = 13.8081 > 0,$$

$$\kappa\rho_b^* = (26 + 4\sqrt{51})H^2 = 54.566H^2.$$

$G(k_*) > 0$: four-dimensional gravity is attractive with the author's weights and $\Lambda$ unchanged. (Without the stabiliser the only static block with $A=0$ is $k_0 = 8$ with $G = -1$, repulsive.) The linearised system has the eigenvalues $0, \pm i\omega_r$,

$$\omega_r^2 = \Big(\frac{31552}{2249} - \frac{2416\sqrt{51}}{6747}\Big)H^2, \qquad \omega_r = 3.38705H,$$

so the block is stable; the radion mass is $m_r = 3.387H = 48.42$ TeV. The whole stabiliser family $p_0 = \nu\rho_b$ ($\nu = 1/4$ to 3) gives stable blocks with $\omega_r = 3.33H$ to $4.10H$ (DERIVED).

## 4.2 The primordial era: the deflating extra times

The full nonlinear equations (all three components, the constraint preserved to $6 \times 10^{-10}$) were integrated from an expanding 3-space and a block larger than the stabilised one ($k_0 < k_*$) that deflates at $h_0 = -H/2, -H/3, -H/10$, for $k_0$ from 1.5 to 20 (DERIVED):

| start $k_0$ | $h_0 = -H/10$ | $h_0 = -H/3$ | $h_0 = -H/2$ |
|---|---|---|---|
| 1.5 | singular at $x_4 = 1.44/H$ | singular at $1.26/H$ | singular at $1.14/H$ |
| 2.3 | singular at $0.33/H$ | singular at $0.31/H$ | no expanding start |
| 2.7 | $k(60/H) = 21.415$ | 21.435 | 21.451 |
| 3 | 21.412 | 21.434 | 21.455 |
| 6 | 21.405 | 21.453 | 21.500 |
| 10 | 21.446 | 21.541 | 21.584 |
| 20 | 21.634 | 21.640 | 21.638 |

Every start above the separatrix $k_c = 2.5301, 2.5991, 2.6983$ (for $h_0 = -H/10, -H/3, -H/2$; just above the unstable static block $k_- = 2.4781$) ends at the stable block: the extra times $x_5, x_6, x_7$ and the hidden direction $x_0$ deflate at rates up to $|h| = 2.63H$ (in the stored runs $k_0 = 3, 5, 10$), by up to $f(k_c)/f_* = 2.917$, reach $k_*$ within $x_4 < 1/H$, and then oscillate about it with the radion frequency ($3.3868H$ measured, $3.3870H$ linear), while 3-space keeps expanding; the oscillation energy redshifts like dust ($A\,x_4 \to 0.667$ to 0.670; 2/3 for dust). The extra times therefore deflate exponentially only in a short primordial era and are then frozen: the masses, all of which depend on $f$, become constant.

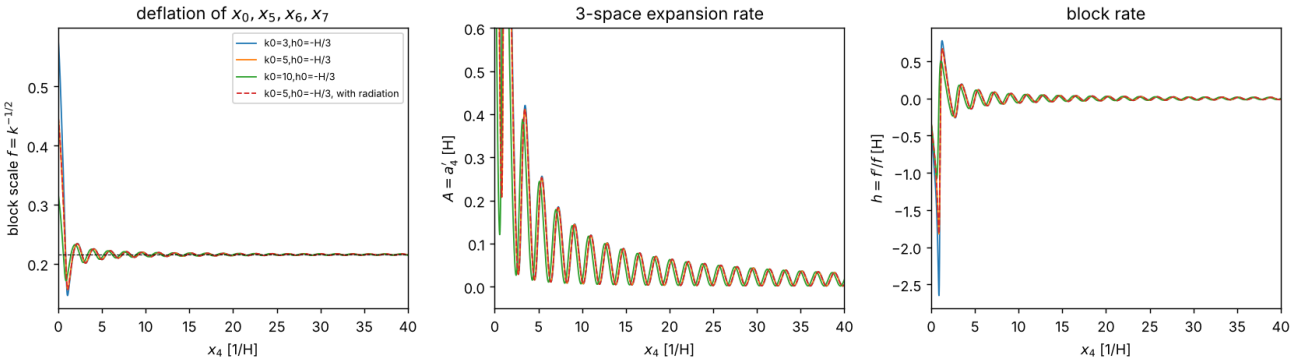



Figure 1: The primordial era: the block scale $f$, the 3-space expansion rate $A$ and the block rate $h$ for starts $k_0 = 3, 5, 10$ at $h_0 = -H/3$ (one with radiation), from the full nonlinear Einstein-Lovelock equations with the author's weights.

## 4.3 The response of the block to matter, and four-dimensional gravity

To first order about the static block with block-averaged matter sources ($\bar{p} = (p_0 + 3p_t)/4$; $A^2$ of first order, $3G(k_*)A^2 = \kappa\delta\rho$) (DERIVED):

$$\delta\ddot{k} + \omega_r^2\,\delta k = -2k_*\big(s_{h3}\,\kappa\delta p_3 + s_{hp}\,\kappa\delta\bar{p} + q_h\,\kappa\delta\rho\big), \qquad s_{h3} = 0.028666, \quad s_{hp} = 0.037320, \quad q_h = -0.009555;$$

the adiabatic shift is $\delta k = (-0.107556\,\kappa\delta p_3 - 0.140028\,\kappa\delta\bar{p} + 0.035852\,\kappa\delta\rho)/H^2$, and 3-space obeys $\delta\dot{A} = -\kappa(\delta\rho + \delta p_3)/(2G(k_*))$. Together with the constraint $3G(k_*)A^2 = \kappa\rho$ and the dimensional reduction ($\sum_p \alpha_p \mathcal{L}_{(p)} - 2\Lambda - 2\kappa\rho_b = G(k)R^{(4)} + \text{topological} - 2G(k)\Lambda_4$, $p$ the Lovelock order), the four-dimensional gravity of UM-2 is general relativity,

$$G^{(4)}_{\mu\nu} + \Lambda_4 g_{\mu\nu} = 8\pi G_N T^{(4)}_{\mu\nu}, \qquad 8\pi G_N = \frac{\kappa}{G(k_*)V_\perp}, \qquad V_\perp = \frac{2f_*^4 v_t(1 - e^{-3\eta})}{3H},$$

with $\Lambda_4 = [6H^2k(1 - 2w_2k) + \Lambda + \kappa\rho_b]/G(k) = 0$ at the static point; the observed dark energy needs $\Lambda_4/H^2 \sim (H_0/H)^2 = 1.0 \times 10^{-92}$ (tuned; OPEN).

# 5 Step 2: the masses from the $x_0$ dependence of the wave functions

Every four-dimensional mass is an eigenvalue of the $x_0$ operator of its field on the static block; the eigenfunctions depend on $x_0$ through $y(6Hx_0)$ and are periodic with period $\pi/(6H)$ (the author's requirement). With $\xi = y/L$:

## 5.1 Gauge bosons and the graviton

$$-e^{-3\eta\xi}\big(e^{3\eta\xi}u'\big)' = \Lambda u, \qquad \Lambda = (m\,fL)^2, \qquad u'(0) = 0 \text{ (even) or } u(0) = 0 \text{ (odd)}, \qquad u'(-1) = \beta u(-1),$$

$$\text{even: } \Big[q^2 + \frac{3\eta}{2}\Big(\beta + \frac{3\eta}{2}\Big)\Big]\sin q - \beta q\cos q = 0, \qquad \text{odd: } q\cos q + \Big(\beta + \frac{3\eta}{2}\Big)\sin q = 0, \qquad q^2 = \Lambda - \frac{9\eta^2}{4}.$$

| mode ($\eta = 3$) | $\Lambda = (m\,fL)^2$ | status |
|---|---|---|
| photon, gluons, graviton ($\beta = 0$, flat profile) | 0 exactly | DERIVED |
| first even Kaluza-Klein level | $9\eta^2/4 + \pi^2 = 30.1196$ | DERIVED (finite elements $1.9 \times 10^{-10}$) |
| first odd level | 27.0890 ($q\cot q = -9/2$) | DERIVED (finite elements $4.5 \times 10^{-10}$) |
| gap | $9\eta^2/4 = 20.25$ | DERIVED |
| $W^+$, $W^-$ ($\beta_W = 0.011912$) | $1.32149 \times 10^{-5}$ | DERIVED (fit, finite elements $3 \times 10^{-14}$) |
| $Z$ ($\beta_Z = \beta_W/\cos^2\theta_W = 0.015343$, angle of the couplings $\cos^2\theta_W = (m_W^2/m_Z^2)/\rho_0 = 0.776368$) | $1.70150 \times 10^{-5}$ | DERIVED |
| Dirichlet bound ($\beta \to \infty$) | 0.010016: $m_W \le 0.01824\, m_{KK}$ | DERIVED |

Here $\theta_W$ is the angle of the gauge couplings, $\cos^2\theta_W = \hat{g}^2/(\hat{g}^2 + \hat{g}'^2)$. The bending of the tip-localised profiles moves the mass ratio $m_W^2/m_Z^2 = \Lambda_W/\Lambda_Z$ away from $\cos^2\theta_W$, and the electroweak parameter $\rho = \Lambda_W/(\Lambda_Z\cos^2\theta_W)$ exceeds 1 ($\rho - 1 = 3.2 \times 10^{-8}$ at $\beta_W = 10^{-6}$: custodial symmetry at leading order; $6.4 \times 10^{-5}$ at $\beta_W = 0.002$, $3.2 \times 10^{-3}$ at 0.1). This tree-level custodial violation is fitted to the measured $\rho_0 - 1 = 0.00038$ (the violation beyond the Standard-Model loops), and the angle of the couplings is fixed by the measured masses, $\rho = m_W^2/(m_Z^2\cos^2\theta_W) = \rho_0$, i.e. $\cos^2\theta_W = (m_W^2/m_Z^2)/\rho_0 = 0.776368$ (on-shell $m_W^2/m_Z^2 = 0.776663$); both masses are then reproduced. With $m_W = 80.3625$ GeV this gives the scales (DERIVED)

| quantity | value |
|---|---|
| Kaluza-Klein mass $m_{KK} = m_W\sqrt{\Lambda_1/\Lambda_W}$ | 121.32 TeV |
| $H = m_{KK} f_* \eta/\sqrt{\Lambda_1}$ | 14.296 TeV |
| proper half cell $fL$; period of $x_0$, $\pi/(6H)$ | $8.926 \times 10^{-21}$ m; $7.227 \times 10^{-21}$ m |
| gap $3H/(2f_*)$ | 99.48 TeV |
| radion mass $3.387H$ | 48.42 TeV |
| model $m_Z = m_W\sqrt{\Lambda_Z/\Lambda_W}$ | 91.1879 GeV (the measured value, reproduced) |

## 5.2 Fermions: masses and mixing of three generations

$$h_j = j\Big[-\frac{i}{f}\sigma_1\partial_y + \mu\,\sigma_2 + e^{-a_4}k\,\sigma_3\Big],$$

with $k$ the momentum of the mode along $x_1$ (not the block variable) and $c = f\mu L$. At $k = 0$ the doublet ($c_L > 0$) and singlet ($c_R < 0$) solutions that satisfy the fold conditions reach the tip with $(a_L, b_L) = (C_L - c_L s_L, -s_L e)$

and $(a_R, b_R) = (s_R e, C_R + c_R s_R)$, $s = \sinh w/w$, $C = \cosh w$, $w = \sqrt{c^2 - e^2}$; with the tip mixing $b_L = \kappa_f b_R$, $a_R = -\kappa_f a_L$ of one generation the mass is the lowest root of

$$D(e) = e^2 s_L s_R - \kappa_f^2 (C_L - c_L s_L)(C_R + c_R s_R) = 0, \qquad e \simeq \kappa_f \zeta(c_L)\zeta(-c_R), \quad \zeta(c)^2 = \frac{c\, e^{-c}}{\sinh c}.$$

For three generations the tip conditions $b_L = \kappa b_R$, $a_R = -\kappa^\dagger a_L$ (vectors over the generations, $\kappa$ the $3 \times 3$ tip matrix of a sector) give, with the doublet amplitudes $\mathcal{A}$ and the singlet amplitudes $\mathcal{B}$,

$$\Big[\mathrm{diag}(b_{L\alpha}) + \kappa\, \mathrm{diag}\Big(\frac{b_{Ri}}{a_{Ri}}\Big)\kappa^\dagger\, \mathrm{diag}(a_{L\alpha})\Big]\mathcal{A} = 0,$$

and for a common doublet bulk mass and a unitary $\kappa$ the determinant is $\prod_i (b_L a_{Ri} + a_L b_{Ri})/a_{Ri}$ with $b_L a_R + a_L b_R = -D(e)|_{\kappa_f = 1}$, the null vectors are $\mathcal{A}^{(i)} = \kappa \hat{e}_i$ and $\mathcal{B}^{(i)} \propto \hat{e}_i$: the masses are the one-generation roots with $\kappa_f = 1$ and the left-handed mixing matrix is the tip matrix (DERIVED, `um2_flavour.py`, mpmath with 50 digits: the roots of the $3 \times 3$ determinant agree with the one-generation roots to $3.0 \times 10^{-42}$ in the four sectors, with $\kappa_\nu = U_{PMNS}$, $\kappa_e = 1$, $\kappa_u = 1$, $\kappa_d = V_{CKM}$, and the null vectors with the columns of $\kappa$ to $1.1 \times 10^{-16}$; for a random complex non-unitary tip matrix the exact roots agree with the singular values of $\zeta(c_L)\, \kappa\, \mathrm{diag}(\zeta(-c_{Ri}))$ to $3.5 \times 10^{-5}$, $2.8 \times 10^{-6}$, $2.0 \times 10^{-6}$ and the null vectors with its left singular vectors).

With one doublet bulk mass per sector, fixed by the heaviest member with $c_R = -c_L$ ($\tau$, $t$), the masses require (DERIVED, `um2_spectra.py`; exact roots; direct shooting of the coupled $(L, R)$ Dirac system agrees with the determinant to $6.1 \times 10^{-10}$ ($e$), $1.5 \times 10^{-11}$ ($\tau$), $5.6 \times 10^{-12}$ ($t$); the first-order formula agrees with the exact root to the relative accuracy in the last column):

| fermion | mass | doublet $c$ | singlet $c_R$ | $\mu_R/H$ | singlet width $fL/(2\lvert c_R\rvert)$ | $e = m\, fL$ | first order |
|---|---|---|---|---|---|---|---|
| $\nu_1$ | 1.000 meV | 5.95288 | −35.04518 | −54.194 | $1.27 \times 10^{-22}$ m | $4.5235 \times 10^{-17}$ | $5.7 \times 10^{-7}$ |
| $\nu_2$ | 8.712 meV | 5.95288 | −32.84810 | −50.796 | $1.36 \times 10^{-22}$ m | $3.9409 \times 10^{-16}$ | $6.1 \times 10^{-7}$ |
| $\nu_3$ | 50.140 meV | 5.95288 | −31.07017 | −48.047 | $1.44 \times 10^{-22}$ m | $2.2681 \times 10^{-15}$ | $6.5 \times 10^{-7}$ |
| $e$ | 0.51100 MeV | 5.95288 | −14.55391 | −22.506 | $3.07 \times 10^{-22}$ m | $2.3115 \times 10^{-8}$ | $1.4 \times 10^{-6}$ |
| $\mu$ | 105.658 MeV | 5.95288 | −8.98093 | −13.888 | $4.97 \times 10^{-22}$ m | $4.7795 \times 10^{-6}$ | $2.2 \times 10^{-6}$ |
| $\tau$ | 1776.93 MeV | 5.95288 | −5.95288 | −9.206 | $7.50 \times 10^{-22}$ m | $8.0380 \times 10^{-5}$ | $6.8 \times 10^{-6}$ |
| $u$ | 2.16 MeV | 3.38230 | −15.43001 | −23.861 | $2.89 \times 10^{-22}$ m | $9.7708 \times 10^{-8}$ | $1.3 \times 10^{-4}$ |
| $c$ | 1.2730 GeV | 3.38230 | −8.76830 | −13.559 | $5.09 \times 10^{-22}$ m | $5.7585 \times 10^{-5}$ | $2.2 \times 10^{-4}$ |
| $t$ | 172.56 GeV | 3.38230 | −3.38230 | −5.230 | $1.32 \times 10^{-21}$ m | $7.8058 \times 10^{-3}$ | $1.1 \times 10^{-3}$ |
| $d$ | 4.70 MeV | 3.38230 | −14.62578 | −22.617 | $3.05 \times 10^{-22}$ m | $2.1261 \times 10^{-7}$ | $1.3 \times 10^{-4}$ |
| $s$ | 93.5 MeV | 3.38230 | −11.51582 | −17.808 | $3.88 \times 10^{-22}$ m | $4.2295 \times 10^{-6}$ | $1.7 \times 10^{-4}$ |
| $b$ | 4.183 GeV | 3.38230 | −7.50053 | −11.599 | $5.95 \times 10^{-22}$ m | $1.8922 \times 10^{-4}$ | $2.6 \times 10^{-4}$ |

(doublets: $\mu_L = 9.2055\, H$, $\mu_Q = 5.2304\, H$; the masses are the PDG values, the light-quark masses in $\overline{MS}$ at 2 GeV, $m_c(m_c)$, $m_b(m_b)$ and the directly measured $m_t$, used as tree-level targets; $m_1$ is ASSUMED, $m_2$, $m_3$ follow from

the NuFIT 6.0 splittings of step 5.) The mixing matrices are $U = R_{23}U_{13}(\delta)R_{12}$ in the standard form

$$U = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix}, \qquad J = c_{12}s_{12}c_{23}s_{23}c_{13}^2 s_{13}\sin\delta,$$

with the NuFIT 6.0 values ($\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{23} = 0.470$, $\sin^2\theta_{13} = 0.02215$, $\delta = 212°$; $J = -0.01777$, $J_{max} = 0.0335$) for the PMNS matrix and the PDG values ($s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta = 1.147$; $J = 3.117 \times 10^{-5}$) for the CKM matrix, both unitary to $1.1 \times 10^{-16}$ (DERIVED). The first even Kaluza-Klein gauge mode couples to the zero modes with the overlaps 0.14434 (every left-handed lepton) and $-0.30066$ (every left-handed quark) relative to the zero mode, and in the mass basis its largest off-diagonal coupling is $1.0 \times 10^{-17}$: no tree-level flavour-changing neutral current (with the three charged-lepton $|c_R|$ used as different doublet localisations it would be 0.033). $\mathrm{BR}(\mu \to e\gamma) = \frac{3\alpha}{32\pi}\left|\sum_i U^*_{\mu i}U_{ei}m_i^2/m_W^2\right|^2 = 2.8 \times 10^{-55}$ (DERIVED).

### 5.3 dirac16complex

With $M = m$, fold condition $\chi_2(0) = 0$ (even) or $\chi_1(0) = 0$ (odd), tip $\chi_2(-L) = 0$ and $c_\psi = f_* mL = 0.646667$, the exact $k = 0$ spectrum is: the fold zero mode ($\varepsilon = 0$, all eight blocks); even bulk levels $\sqrt{m^2 + (n\pi/(fL))^2}$, the lowest $4.959981\,m$; odd levels $\sqrt{m^2 + p^2}$ with $\tan(p\,fL) = -p/m$, the lowest $3.102204\,m$ ($p = 2.93661\,m$, $p\,fL = 1.89901$); and the fold band $\varepsilon = e^{-a_4}|k|$ (massless, $\varepsilon/|k| = 1$ to $6 \times 10^{-14}$) (DERIVED).

## 6 Step 3: Kohn-Sham states of dirac16complex coupled to all fields

### 6.1 The Kohn-Sham equations

$$h_j\chi_o = \varepsilon_o\chi_o, \qquad M_{eff} = m + \frac{15}{16}\lambda S + \frac{\langle\mathcal{F}\rangle}{\Lambda_\star^3}, \qquad v = -\frac{\lambda}{16}n + \frac{c_V}{\Lambda_\star^6}\sum_{i=1}^{3}\sum_{\psi} q_\psi n_{\psi i},$$

$$n = \sum wgf_o n_o, \quad S = \sum wgf_o s_o, \quad n_o = P\chi_o^\dagger\chi_o, \quad s_o = P\chi_o^\dagger j\sigma_2\chi_o, \quad t_o = P\chi_o^\dagger j\sigma_3\chi_o, \quad P = \frac{e^{-3Hy}}{\mathrm{Vol}_7},$$

$$E_{KS} = \sum gf_o\varepsilon_o - 2\mathrm{Vol}_7\int e^{3Hy}e_{int}dy, \qquad e_{int} = \frac{15}{32}\lambda S^2 - \frac{1}{32}\lambda n^2,$$

$$\rho = \sum wgf_o\varepsilon_o n_o - e_{int}, \qquad p_3 = \sum wgf_o\tfrac{e^{-a_4|k|}}{3}t_o + e_{int}, \qquad p_t = e_{int},$$

$$p_0 = \sum wgf_o[(\varepsilon_o - v)n_o - M_{eff}s_o - e^{-a_4}|k|t_o] + e_{int},$$

with $w = 1/2$ per fold parity, $g = 4r_3(|\vec{n}|^2)$ for the momentum shell $\vec{k} = \Delta k\,\vec{n}$ ($\vec{n} \in \mathbb{Z}^3$, $r_3(|\vec{n}|^2)$ the number of representations of $|\vec{n}|^2$ as a sum of three squares, $\Delta k = 0.25$; here $k$ is the momentum, not the block variable), occupations $f_o$, and $\mathrm{Vol}_7 = (2\pi/\Delta k)^3 v_t f_*^4 = 34.27$ ($v_t = 1$, $a_4 = 0$; the proper seven-volume of a cell is $2\,\mathrm{Vol}_7(1 - e^{-3\eta})/(3H)$). The orbitals are expanded in the exact free eigenfunctions of each sector (60, 120, 240 per sector), iterated to self-consistency (tolerance $10^{-13}$, safeguarded Anderson mixing), and extrapolated in the basis size (Richardson).

### 6.2 Validation

The independent solver reproduces the record's two reference states (at $f = 1$, $a_4 = 0$): $N = 8$, $\lambda = 157.7$: $E_{KS} = -0.037786682923$ (reference $-0.037786682923$), HOMO and LUMO to $10^{-14}$, integrated $p_0 = -0.08860537545$ (reference $-0.08860537548$); $N = 592$, $\lambda = 2.131$: $E_{KS} = 358.30104644787$ (identical), integrated $p_0 = -6.7691415097$ (reference $-6.7691415108$) (DERIVED).

### 6.3 Ground states on the stabilised block

Two families of states (the eight $k = 0$ fold zero modes, $N = 8$; a fold-band gas, $N = 592$) with $\lambda = 0$, the benchmark $\pm\lambda_B$ ($\lambda_B = 7 \times 10^{-5} H^{-6}$: mean fields below 5 % of $m$) and the record's coupling scaled to the block (stress test, $\lambda = 157.7 f_*^4 = 0.3405$ and $2.131 f_*^4 = 0.00460$) (DERIVED; 120 functions per sector, Richardson errors below $1.5 \times 10^{-9}$):

| state | $E_{KS}$ | $E_{int}$ | HOMO | LUMO | iterations |
|---|---|---|---|---|---|
| $N = 8$, $\lambda = 0$ | 0 | 0 | 0 | 0.25 | 1 |
| $N = 8$, $+\lambda_B$ | $-2.053 \times 10^{-4}$ | $-2.053 \times 10^{-4}$ | $-5.13 \times 10^{-5}$ | 0.24995 | 14 |
| $N = 8$, $-\lambda_B$ | $2.053 \times 10^{-4}$ | $2.053 \times 10^{-4}$ | $5.13 \times 10^{-5}$ | 0.25005 | 14 |
| $N = 8$, stress | $-0.70123$ | $-0.42170$ | $-0.14037$ | 0.10859 | 61 |
| $N = 592$, $\lambda = 0$ | 361.11614 | 0 | 0.79057 | 0.82916 | 1 |
| $N = 592$, $+\lambda_B$ | 359.99060 | $-1.12646$ | 0.78676 | 0.82535 | 14 |
| $N = 592$, $-\lambda_B$ | 362.23906 | 1.12121 | 0.79436 | 0.83295 | 13 |
| $N = 592$, stress | 307.67139 | $-32.2217$ | 0.64506 | 0.68347 | 56 |

| state | $\int p_3$ | $\int p_t$ | $\int p_0$ | max $\lvert\delta M\rvert$ | max $\lvert v\rvert$ |
|---|---|---|---|---|---|
| $N = 8$, $+\lambda_B$ | $-2.05 \times 10^{-4}$ | $-2.05 \times 10^{-4}$ | $-4.447 \times 10^{-4}$ | $2.4 \times 10^{-7}$ | $6.7 \times 10^{-4}$ |
| $N = 8$, $-\lambda_B$ | $2.05 \times 10^{-4}$ | $2.05 \times 10^{-4}$ | $4.447 \times 10^{-4}$ | $2.4 \times 10^{-7}$ | $6.7 \times 10^{-4}$ |
| $N = 8$, stress | $-0.4217$ | $-0.4217$ | $-1.30076$ | 1.98 | 1.83 |
| $N = 592$, $\lambda = 0$ | 120.372 | 0 | $8.6 \times 10^{-12}$ | 0 | 0 |
| $N = 592$, $+\lambda_B$ | 119.245 | $-1.1265$ | $-2.43688$ | $1.3 \times 10^{-3}$ | 0.050 |
| $N = 592$, $-\lambda_B$ | 121.493 | 1.1212 | 2.43174 | $1.3 \times 10^{-3}$ | 0.050 |
| $N = 592$, stress | 87.614 | $-32.222$ | $-97.8188$ | 2.09 | 1.88 |

(The integrals are $2\mathrm{Vol}_7 \int e^{3Hy}(\cdot)\, dy$; the $N = 8$, $\lambda = 0$ state has all of them zero.)

Checks (DERIVED): $E_{KS} = 2\mathrm{Vol}_7 \int e^{3Hy} \rho\, dy$ to $4 \times 10^{-15}$; the $y$-conservation law $[e^{3Hy} p_0]_{-L}^{0} = 3H \int e^{3Hy} p_t dy$ to $9 \times 10^{-8}$ (Richardson; the raw residual falls like $n_b^{-3}$); observed basis orders 4.95 (energy) and 2.97 (hidden pressure); the free gas has $w = \int p_3 / \int \rho = 1/3$ exactly (four-dimensional radiation). The self-interaction shifts the fold-band levels almost rigidly (the spacing between $k = 0$ and the first shell is 0.25 free, 0.25000 at $\lambda_B$, 0.2490 in the $N = 8$ stress state). A scalar mean field cannot gap the fold band at $k = 0$: $\chi = (e^{f \int M_{eff} dy}, 0)$ solves $j[-\frac{i}{f}\sigma_1 \partial_y + M_{eff}\sigma_2]\chi = 0$ with both boundary conditions for every $M_{eff}(y)$.

### 6.4 Dark-matter quanta

Eight quanta in the lowest odd level at $k = 0$ on the $N = 8$ ground state add exactly $8\varepsilon_{DM} = 24.817633063$ (computed 24.817633063) at $\lambda = 0$ and 24.81833 at $+\lambda_B$; they are cold, $w = p_3/\rho = 0$ ($2.8 \times 10^{-5}$ with $\lambda_B$); their hidden pressure is $p_0/\rho = 0.772034343$, equal to $-d \ln \varepsilon_{DM} / d \ln f = 0.772034343$ obtained independently from the $x_0$ eigenvalue (DERIVED): the dark-matter mass is an $x_0$ eigenvalue, and it pushes on the block.

### 6.5 The Standard-Model fields acting on dirac16complex

The portal mean fields were applied with amplitude $e$ and shapes given by the sources: for I11 the proper densities of the zero modes times $P = e^{-3Hy}/\mathrm{Vol}_7$, with the density of an unpolarised fermion the average of its doublet and singlet zero-mode densities, $\varrho_c(y) = (2c/L)\, e^{2cy/L}/(1 - e^{-2c})$ for the localisation $c$: the electron $\frac{1}{2}(\varrho_{c_L} + \varrho_{-c_R(e)})$ ($c_L = 5.9529$, $-c_R(e) = 14.5539$), the up quark $\frac{1}{2}(\varrho_{c_Q} + \varrho_{-c_R(u)})$ ($c_Q = 3.3823$, $-c_R(u) = 15.4300$) and the neutrinos, produced left-chiral, $\varrho_{c_L}$; for I12 a uniform gluon condensate. Linear responses of the lowest odd level (DERIVED; equal to the Hellmann-Feynman expectation $d\varepsilon/de = \langle\chi|\partial h/\partial e|\chi\rangle$ to $10^{-8}$):

| portal mean field | $d\varepsilon_{odd}/de$ | change of the fold-band gap |
|---|---|---|
| I11, electron density ($v$) | 0.824779 | $5.4 \times 10^{-11}$ |
| I11, up-quark density ($v$) | 4.328803 | $3.1 \times 10^{-9}$ |
| I11, neutrino density ($v$, lepton doublet zero mode) | 1.252406 | $1.1 \times 10^{-10}$ |
| I12, gluon condensate (uniform $M$) | 0.707196 | $2.9 \times 10^{-11}$ |

The broad quark doublet zero mode ($c_Q = 3.38$) overlaps the dark-matter orbital most. The fold band stays massless under every portal. Every orbital has $\chi_2(-L) = 0$, so the scalar density $\bar\Psi\Psi = 2s_2ab$ vanishes at the tip: no tip operator built on $\bar\Psi\Psi$ reaches dirac16complex; the upper component does not vanish there, so a chiral tip current would reach $\Psi_A$ (the projector $P_+ = (1 - B\gamma^{x_0})/2$ annihilates every $\Psi_B$ orbital at the tip), but UM-2 contains none (step 4). The I12 response defines the four-dimensional coefficient of the dark-matter operator: $G_i = \mathcal{O}\, c_i g_i^2/\Lambda_\star^3$ with $\mathcal{O} = 0.7072$.

### 6.6 Back-reaction on the block

The block-averaged hidden pressure $\bar p = (p_0 + 3p_t)/4$ of a Kohn-Sham state displaces the block by $\delta k = \kappa\bar p/(3H^2 - 4\kappa\rho_b^*/k_*) = -0.140028\,\kappa\bar p/H^2$ (stiffness $-7.1414H^2$), without changing the four-dimensional cosmological term at linear order ($\frac{d}{dk}[6H^2k(1 - 2w_2k) + \kappa\rho_b] = -6 \times 10^{-12}$ at $k_*$). Cell averages (units of the Kohn-Sham calculation, $H = 1$): $\bar p = -1.16 \times 10^{-5}$ ($N = 8$, $+\lambda_B$), $-0.0636$ ($N = 592$, $+\lambda_B$), $\approx 0$ (free gas); the shift is negligible for any realistic density (relative size $(A/H)^2$). The difference $p_0 - p_t$ ($-0.0574$ for $N = 592$, $+\lambda_B$) drives a block-shape mode outside the order-0 metric (OPEN).

## 7 Step 4: the electroweak sector

**Charges and couplings** (DERIVED): in every generation $Q = T_3 + Y$ gives the up-type quarks $u$, $c$, $t$ the charge $2/3$, the down-type quarks $d$, $s$, $b$ the charge $-1/3$, the neutrinos 0 and the charged leptons $e$, $\mu$, $\tau$ the charge $-1$, for both chiralities; $[T_3, T_\pm] = \pm T_\pm$ with $Y(W) = 0$: $W^+ = (W^1 - iW^2)/\sqrt{2}$ **carries** $+e$**,** $W^- = (W^1 + iW^2)/\sqrt{2}$ **carries** $-e$, and $W^-$ is the antiparticle of $W^+$; the photon couples with $e = gg'/\sqrt{g^2 + g'^2}$ times $Q$, the $Z$ with $\sqrt{g^2 + g'^2}(T_3 - Q\sin^2\theta_W)$.

**Anomalies** (DERIVED, exact fractions). With every zero mode written as a left-handed Weyl field ($-Y$ for a right-handed one), $n_c$ colours and $n_w$ weak components, one generation gives

$$\mathcal{A}_{SU(3)^2U(1)} = \sum_{\text{triplets}} n_w Y = 0, \qquad \mathcal{A}_{SU(2)^2U(1)} = \sum_{\text{doublets}} n_c Y = \tfrac{1}{2} - \tfrac{1}{2} = 0,$$

$$\mathcal{A}_{U(1)^3} = \sum n_c n_w Y^3 = \tfrac{1}{36} - \tfrac{8}{9} + \tfrac{1}{9} - \tfrac{1}{4} + 1 = 0,$$

$\mathcal{A}_{grav^2U(1)} = \sum n_c n_w Y = 1 - 2 + 1 - 1 + 1 = 0$ and $\mathcal{A}_{SU(3)^3} = 2 - 1 - 1 = 0$; three generations have twelve $SU(2)$ doublets (no Witten anomaly). Three lepton generations with a single quark generation would leave $\mathcal{A}_{SU(2)^2U(1)} = -1$ and $\mathcal{A}_{U(1)^3} = 3/2$.

**The two halves of dirac16complex.** $B$ commutes with the four-dimensional Lorentz generators and with $C$, so $\Psi = \Psi_A + \Psi_B$ ($B = -1$, $B = +1$) splits into Lorentz-invariant halves and $\bar\Psi\Psi = \bar\Psi_A\Psi_A + \bar\Psi_B\Psi_B$; the identity $\Gamma_4 = -B\gamma^{x_0}$ (DERIVED, $16 \times 16$) gives $P_+ = (1 + \Gamma_4)/2 = (1 - B\gamma^{x_0})/2$, which equals $(1 + \gamma^{x_0})/2$ on $\Psi_A$ and $(1 - \gamma^{x_0})/2$ on $\Psi_B$; the latter annihilates every orbital at the tip. The chiral charge densities $u^\dagger P_+ u$ at the tip (what a chiral tip current would couple to) are (DERIVED): $\Psi_A$ fold band $a_0(-L)^2 = 0.163$ per unit length in each of its four blocks, $\Psi_B$ fold band 0; $\Psi_A$ odd (heavy) states 0.257, $\Psi_B$ odd states 0; every transition current $\bar\Psi_A\gamma^\mu P_+\Psi_B$ vanishes at the tip.

**The $Z$ invisible width and the tip $Z$ portal of version 1.** Version 1 coupled the $\Psi_A$ fold band (two Weyl species) to the $Z$ through the chiral tip portal $S_{I9} = \int_{y=-L} d^7x\sqrt{|g_{tip}|}\,(c_Z + c'_Z\varsigma)(i\bar\Psi\gamma^\mu P_+\Psi)\,\frac{iv_\Sigma^2}{2}\mathrm{tr}(\Sigma^\dagger D_\mu\Sigma\tau^3)$ ($\varsigma = \pm 1$ the side of the tip). With three neutrinos and a fold-band coupling $r$ times a neutrino's,

$$N_\nu = \frac{\Gamma_{inv}}{\Gamma(Z \to \nu\bar\nu)} = 3 + 2r^2;$$

the measured $N_\nu = 2.9963 \pm 0.0074$ (PDG 2025) is reproduced by $r = 0$ (0.5 standard deviations) and bounds $r^2 < 0.0056$ (two standard deviations). Even at that bound the neutral-current rate of the fold band, $r^2$ times the neutrino's ($\propto r^2 G_F^2 T^5$ against $H \propto T^2/M_{Pl}$), keeps it in equilibrium down to $T \approx 2.6\,\mathrm{MeV} \times r^{-2/3} = 14.7$ MeV, where $g_s = 10.75$, so it would carry the neutrino temperature and add

$$\Delta N_{eff} = n_W \Big(\frac{10.75}{g_s}\Big)^{4/3} F, \qquad F = \Big[\frac{T_\nu}{T_\gamma}\Big(\frac{11}{4}\Big)^{1/3} \Big/ (1 + h_\nu)\Big]^4 = 1.0095$$

($n_W = 2$ Weyl species; $F$ corrects for the non-instantaneous neutrino decoupling of step 5, $T_\nu/T_\gamma = 0.71664$ and the comoving heating $h_\nu = 0.00165$ with one common temperature: a species decoupled earlier is not heated by $e^+e^-$ annihilation), $\Delta N_{eff} = 2.02$, excluded by $N_{eff} = 2.99 \pm 0.17$ (Planck 2018) and $2.86 \pm 0.13$ (ACT DR6). The oriented part $c'_Z$ (odd under the fold reflection) would let the heavy odd $\Psi_A$ states decay, $\Psi_A^{odd} \to \Psi_A^{fold} + Z$, and through the inverse decays and the exchange process $\Psi_A^{fold} Z \to \Psi_A^{fold} Z$ keep the fold band in equilibrium until the odd states disappear, $T \sim M_{DM}/30$ (above the electroweak scale, $g_s = 105.75$: the Standard Model without a Higgs particle), or, for $c'_Z$ of the size of a neutrino's coupling, until the $Z$ bosons annihilate near 15 GeV ($g_s = 86.25$: $t$, $W$, $Z$ gone) (ESTIMATE): then $\Delta N_{eff} = 0.0958$ to 0.1257 and $N_{eff} = 3.139$ to 3.169, 2.1 to 2.4 standard deviations above ACT DR6 (the three-flavour value 3.0432: 1.4). UM-2 therefore drops the portal, $c_Z = c'_Z = 0$ (DERIVED, `um2_electroweak.py`): $N_\nu = 3$, neither fold band is ever thermalised, and both halves keep an exact fold parity. (For reference, the plasma counts used are $g_s = 10.75$: photons, $e^\pm$, three neutrinos; 61.75 just above the QCD crossover: + gluons, $u$, $d$, $s$, $\mu$; 86.25 below the top, $W$ and $Z$ masses; 105.75 above the electroweak scale. Right-handed neutrinos are never thermalised.)

**The $Z$ widths of three generations** (tree level, massless fermions; $\Gamma(Z \to f\bar{f}) = N_c G_F m_Z^3 (g_V^2 + g_A^2)/(6\sqrt{2}\pi)$, $g_V = T_3 - 2Q \sin^2\theta_W$ with the $\overline{MS}$ value $\sin^2\theta_W = 0.23122$, $g_A = T_3$; DERIVED):

| channel | each $\nu$ | invisible ($3\nu$) | each of $e$, $\mu$, $\tau$ | each of $u$, $c$ | each of $d$, $s$, $b$ | hadronic | total |
|---|---|---|---|---|---|---|---|
| tree level (MeV) | 165.9 | 497.7 | 83.4 | 285.4 | 367.9 | 1674.4 | 2422.3 |
| with $1 + \alpha_s/\pi$ on the quarks (MeV) | | | | | | 1737.3 | 2485.2 |
| measured (PDG 2025, MeV) | | 499.1 | | | | | 2495.5 |

The top quark is too heavy for the $Z$ to decay into it; $\alpha_s(m_Z) = 0.1180$. The ratio $R_\ell = \Gamma_{had}/\Gamma_{\ell\ell} = 20.83$ compares with the measured 20.77. The remaining differences (0.4 % in the total) are the size of the electroweak loop corrections, which UM-2 does not compute; with the tree-level angle of the couplings of the mass fit, $\sin^2\theta_W = 0.22363$, the total changes by +0.82 % (a per-cent scheme ambiguity). The one-generation conflict of version 1 (a total of 1232 MeV) is gone.

# 8 Step 5: the flavour conversion of the three neutrinos

## 8.1 Masses, mixing and the equations

The neutrino masses are the $x_0$ eigenvalues of step 2 with the lightest mass chosen (ASSUMED) and the normal ordering (the NuFIT 6.0 best fit; the inverted ordering has $\Delta\chi^2 = 6.1$):

$$m_1 = 1.000 \text{ meV}, \qquad m_2 = \sqrt{m_1^2 + \Delta m_{21}^2} = 8.712 \text{ meV}, \qquad m_3 = \sqrt{m_1^2 + \Delta m_{31}^2} = 50.140 \text{ meV},$$

$\Delta m_{21}^2 = 7.49 \times 10^{-5}$ eV$^2$, $\Delta m_{31}^2 = 2.513 \times 10^{-3}$ eV$^2$; $\sum m_\nu = 0.0599$ eV, below the DESI DR2 bound 0.064 eV (DERIVED, `um2_neutrino.py`, 15 checks). The PMNS matrix is the neutrino tip matrix of step 2. The flavour

amplitudes $\psi_\alpha$ of a neutrino of energy $E$ evolve with

$$i\frac{d\psi}{dt} = \mathcal{H}\psi, \qquad \mathcal{H} = \frac{1}{2E}\,U\,\mathrm{diag}(m_1^2, m_2^2, m_3^2)\,U^\dagger + \mathrm{diag}(V_{CC}, 0, 0), \qquad V_{CC} = \sqrt{2}\,G_F n_e$$

(the flavour-universal neutral-current potential drops out; antineutrinos: $U \to U^*$, $V_{CC} \to -V_{CC}$), solved exactly by $S = e^{-i\mathcal{H}L}$ for a constant density and by the ordered product of layer operators otherwise, with $P(\nu_\alpha \to \nu_\beta) = |S_{\beta\alpha}|^2$. In vacuum

$$P(\nu_\alpha \to \nu_\beta) = \Big|\sum_i U_{\beta i} U_{\alpha i}^* e^{-i m_i^2 L/(2E)}\Big|^2,$$

$$P(\nu_\mu \to \nu_e) - P(\nu_e \to \nu_\mu) = -16 J \sin\Delta_{21}\sin\Delta_{31}\sin\Delta_{32}, \qquad \Delta_{ij} = \frac{\Delta m_{ij}^2 L}{4E}.$$

## 8.2 Vacuum, matter and accelerators

Checks (DERIVED): the exact evolution reproduces the vacuum formula to $3.9 \times 10^{-15}$, unitarity to $1.8 \times 10^{-15}$, CPT ($P(\nu_\alpha \to \nu_\beta) = \bar{P}(\bar\nu_\beta \to \bar\nu_\alpha)$) to $2.8 \times 10^{-16}$ and the T-odd difference to $5.9 \times 10^{-14}$ at reactor, accelerator and atmospheric baselines. Reactor $\bar\nu_e$ of 4 MeV survive with $\bar{P}_{ee} = 0.9183$ at 1.65 km, 0.2263 at 52.5 km and 0.2674 at 180 km; the vacuum $\nu_\mu$ survival has its first minimum, 0.0060, at $L/E = 1516$ km/GeV. The 1-3 MSW resonance, $2EV_{CC} = \Delta m_{ee}^2 \cos 2\theta_{13}$ with $\Delta m_{ee}^2 = c_{12}^2\Delta m_{31}^2 + s_{12}^2\Delta m_{32}^2$, lies at 11.135 GeV for $\rho = 2.8$ g/cm$^3$, $Y_e = 0.5$ ($n_e = \rho Y_e N_A$); the exact three-flavour Hamiltonian has $|U_{e3}^m|^2 = 1/2$ at 11.133 GeV (agreement $1.6 \times 10^{-4}$; with $\Delta m_{31}^2$ alone the formula gives 11.24 GeV). The 1-2 resonance at the centre of the Sun ($n_e = 100 N_A$ cm$^{-3}$), $2EV_{CC}c_{13}^2 = \Delta m_{21}^2 \cos 2\theta_{12}$, lies at 1.93 MeV. Long-baseline appearance in matter of constant density ($\delta = 212°$, normal ordering):

| experiment | $L$ (km) | $E$ (GeV) | $\rho$ (g/cm$^3$) | $P(\nu_\mu \to \nu_e)$ | $\bar{P}(\bar\nu_\mu \to \bar\nu_e)$ | vacuum $P(\nu_\mu \to \nu_e)$ | $P(\nu_\mu \to \nu_\mu)$ |
|---|---|---|---|---|---|---|---|
| T2K | 295 | 0.6 | 2.6 | 0.0517 | 0.0313 | 0.0478 | 0.0085 |
| NOvA | 810 | 2.0 | 2.84 | 0.0481 | 0.0236 | 0.0404 | 0.1010 |
| DUNE | 1300 | 2.5 | 2.848 | 0.0701 | 0.0195 | 0.0491 | 0.0098 |

The matter effect raises the neutrino and lowers the antineutrino appearance for the normal ordering, and the phase $\delta = 212°$ adds to the asymmetry.

## 8.3 The Sun

With the electron density $n_e(r) = N_A \min(245\, e^{-10.54\, r/R_\odot}, 100)$ cm$^{-3}$ (ASSUMED profile) and production at the radius $r_0$, the adiabatic survival probability is

$$P_{ee} = c_{13}^4\Big(\frac{1}{2} + \frac{1}{2}\cos 2\theta_{12}\cos 2\theta_{12}^m(r_0)\Big) + s_{13}^4, \qquad \cos 2\theta_{12}^m = \frac{\cos 2\theta_{12} - A}{\sqrt{(\cos 2\theta_{12} - A)^2 + \sin^2 2\theta_{12}}}, \qquad A = \frac{2EV_{CC}(r_0)c_{13}^2}{\Delta m_{21}^2},$$

and the numerical three-flavour evolution through 20000 and 40000 layers from $r_0$ to the surface, followed by the incoherent sum over the mass states, $P_{ee} = \sum_i |U_{ei}|^2 P(\nu_i)$, gives (DERIVED):

| source | $E$ (MeV) | $r_0/R_\odot$ | $P_{ee}$ adiabatic | $P_{ee}$ numerical |
|---|---|---|---|---|
| pp | 0.30 | 0.10 | 0.5409 | 0.5408 |
| $^7$Be | 0.862 | 0.06 | 0.5197 | 0.5196 |
| pep | 1.44 | 0.07 | 0.4978 | 0.4976 |
| $^8$B | 5.0 | 0.05 | 0.3771 | 0.3767 |
| $^8$B | 10.0 | 0.05 | 0.3194 | 0.3185 |
| $^8$B | 15.0 | 0.05 | 0.3055 | 0.3043 |

The two agree to $1.3 \times 10^{-3}$ (the layer convergence is $2.2 \times 10^{-6}$): the conversion is adiabatic, vacuum dominated at low energy ($P_{ee} \to c_{13}^4(1 - \frac{1}{2}\sin^2 2\theta_{12}) + s_{13}^4$) and matter dominated for $^8$B ($P_{ee} \to c_{13}^4 \sin^2\theta_{12} + s_{13}^4$).

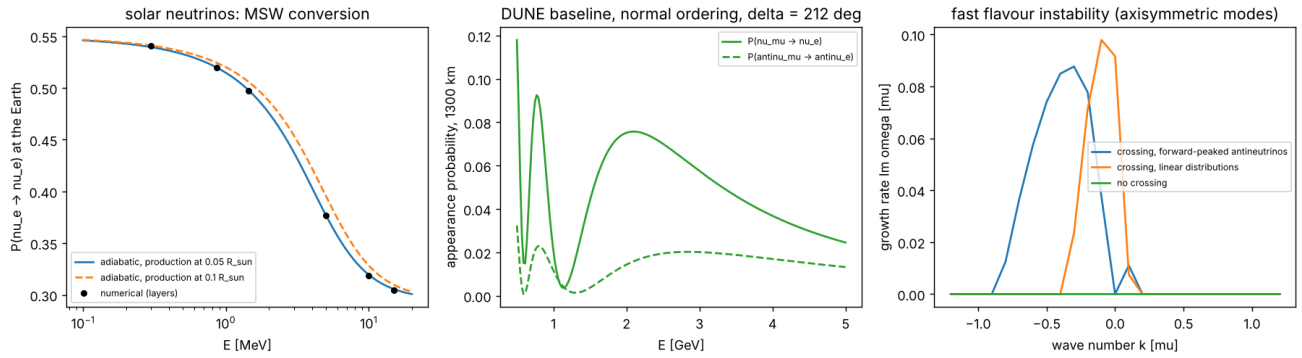



Figure 2: Flavour conversion in UM-2. Left: the solar $\nu_e$ survival probability (adiabatic formula for two production radii, numerical evolution as points). Middle: $\nu_\mu \to \nu_e$ and $\bar\nu_\mu \to \bar\nu_e$ appearance over 1300 km of rock (normal ordering, $\delta = 212°$). Right: the growth rate of the fast flavour instability against the wave number for the three angular distributions of the text.

## 8.4 The early Universe and $N_{eff}$

The thermal potential of the $e^\pm$ plasma,

$$V_e - V_\mu = -\frac{8\sqrt{2}\,G_F p}{3m_W^2}\,\frac{7\pi^2}{60}T^4,$$

falls below $\Delta m^2 \cos 2\theta/(2p)$ at the mean momentum $p = 3.15T$ when $T = 4.99$ MeV for the atmospheric and $T = 2.39$ MeV for the solar splitting (DERIVED; $m_W$ of the UM-2 fit): the flavours mix while the neutrinos decouple. The decoupling is computed from the temperature equations (Escudero's method; MeV units, $H = \sqrt{8\pi\rho/3}/M_{Pl}$)

$$\frac{dT_\gamma}{dt} = -\frac{4H\rho_\gamma + 3H(\rho_e + p_e) + 3HT_\gamma \frac{dP_{int}}{dT_\gamma} + \frac{\delta\rho_{\nu_e}}{\delta t} + 2\frac{\delta\rho_{\nu_\mu}}{\delta t}}{\frac{d\rho_\gamma}{dT_\gamma} + \frac{d\rho_e}{dT_\gamma} + T_\gamma \frac{d^2P_{int}}{dT_\gamma^2}}, \qquad \frac{dT_{\nu_\alpha}}{dt} = -HT_{\nu_\alpha} + \frac{\delta\rho_{\nu_\alpha}/\delta t}{d\rho_{\nu_\alpha}/dT_{\nu_\alpha}},$$

$$\frac{\delta\rho_{\nu_e}}{\delta t} = \frac{G_F^2}{\pi^5}\Big[4(g_{eL}^2 + g_{eR}^2)F(T_\gamma, T_{\nu_e}) + 2F(T_{\nu_\mu}, T_{\nu_e})\Big], \qquad \frac{\delta\rho_{\nu_\mu}}{\delta t} = \frac{G_F^2}{\pi^5}\Big[4(g_{\mu L}^2 + g_{\mu R}^2)F(T_\gamma, T_{\nu_\mu}) - F(T_{\nu_\mu}, T_{\nu_e})\Big],$$

$$F(T_1, T_2) = 32 f_a (T_1^9 - T_2^9) + 56 f_s T_1^4 T_2^4 (T_1 - T_2), \qquad f_a = 0.884, \quad f_s = 0.829,$$

with $\rho_\gamma = \pi^2 T_\gamma^4/15$, $\rho_{\nu_\alpha} = 7\pi^2 T_{\nu_\alpha}^4/120$ per flavour ($\nu_\tau$ as $\nu_\mu$), the $e^\pm$ density and pressure with the electron mass, the effective couplings $g_{eL} = 0.727$, $g_{eR} = 0.233$, $g_{\mu L} = -0.273$, $g_{\mu R} = 0.233$ (or the UM-2 tree-level couplings $\frac{1}{2} + s^2$, $s^2$, $-\frac{1}{2} + s^2$, $s^2$, $s^2 = 0.22363$), and the QED interaction pressure $P_{int} = P_2 + P_3$,

$$P_2 = -\frac{e^2T^2}{12\pi^2}J_1 - \frac{e^2}{8\pi^4}J_1^2 + \frac{e^2 m_e^2}{16\pi^4}J_2, \qquad P_3 = \frac{e^3 T}{12\pi^4}I^{3/2}, \qquad \rho_{int} = -P_{int} + T\frac{dP_{int}}{dT},$$

$$J_1 = \int_0^\infty dp\,\frac{p^2}{E}n_D, \qquad J_2 = \int_0^\infty\!\!\int_0^\infty dp\,dq\,\frac{pq}{E_p E_q}\ln\left|\frac{p+q}{p-q}\right| n_D(p) n_D(q), \qquad I = \int_0^\infty dp\,\frac{p^2 + E^2}{E}n_D,$$

$n_D = 2/(e^{E/T} + 1)$, $E = \sqrt{p^2 + m_e^2}$. The integrals are evaluated with Gauss-Legendre panels; the logarithmic singularity of $J_2$ at $q = p$ is removed by the substitution $q = p(1 - s^2)$, and the massless limits $P_2 = -5e^2T^4/288$, $P_3 = e^3T^4/(36\sqrt{3}\pi)$, $\rho_e = 7\pi^2T^4/60$ are reproduced to $10^{-16}$. The equations run from $T = 20$ MeV to 5 keV (LSODA, relative tolerance $10^{-11}$), and

$$N_{eff} = \left(\frac{11}{4}\right)^{4/3}\left[\left(\frac{T_{\nu_e}}{T_\gamma}\right)^4 + 2\left(\frac{T_{\nu_\mu}}{T_\gamma}\right)^4\right];$$

for instantaneous decoupling at $T_d = 20$ MeV the plasma entropy $a^3(\rho + p)/T$ (with $\rho_{int}$, $P_{int}$) is conserved and $T_\nu \propto 1/a$. Results (DERIVED):

| run | $N_{eff}$ | published (Escudero 2020, same method, Table 1 there) |
|---|---|---|
| instantaneous decoupling, no QED | 3.0002 | 3.000 |
| instantaneous, $O(e^2)$ QED | 3.0108 | 3.011 |
| instantaneous, $O(e^2)+O(e^3)$ | 3.0098 | 3.010 |
| Fermi-Dirac energy transfer, one neutrino temperature, no QED | 3.0393 | 3.039 |
| the same with $O(e^2)$ | 3.0495 | |
| the same with $O(e^2)+O(e^3)$ | 3.0485 | 3.049 |
| UM-2, separate $\nu_e$ and $\nu_{\mu,\tau}$ temperatures (no oscillations), $O(e^3)$ | 3.0476 | 3.048 (separate temperatures) |
| UM-2, separate temperatures, UM-2 tree-level couplings | 3.0472 | |
| UM-2, one common temperature (efficient oscillations) | 3.0485 | |

With separate temperatures $T_{\nu_e}/T_\gamma = 0.71732$ and $T_{\nu_\mu}/T_\gamma = 0.71621$; $e^+e^-$ annihilation raises the comoving neutrino temperatures by 0.265 % ($\nu_e$, which also has charged-current interactions with the electrons) and 0.109 % ($\nu_{\mu,\tau}$). The plasma of UM-2 at MeV temperatures is that of the three-flavour Standard Model and contains no extra radiation, so the precision Standard-Model value with oscillations and the electron mass in the rates is adopted:

$$N_{eff} = 3.0432 \pm 0.0002 \qquad \text{(Cielo, Escudero, Mangano and Pisanti 2023)},$$

which the computed range 3.0476 to 3.0485 approaches to 0.005 (the remaining difference is the electron mass in the rates and the momentum dependence, which the method of temperatures does not resolve).

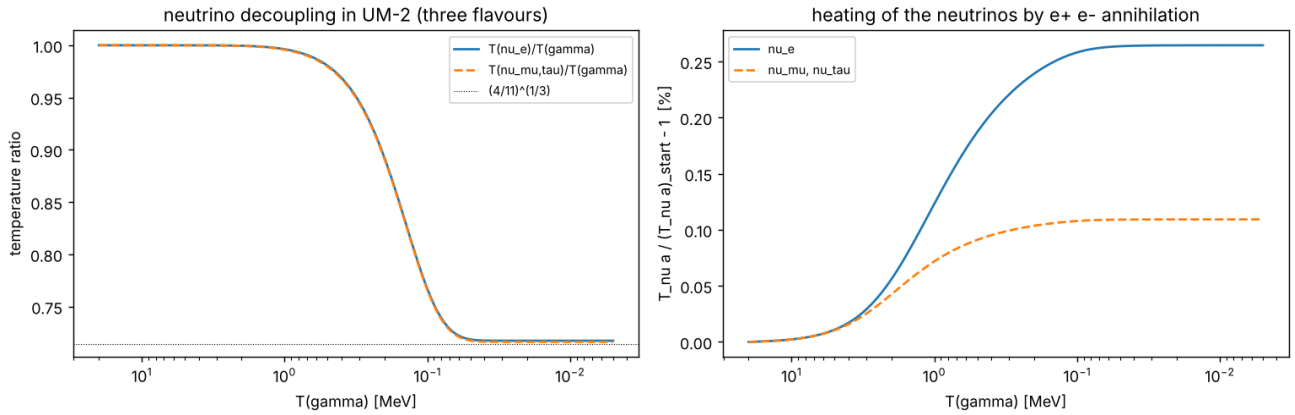



Figure 3: Neutrino decoupling in UM-2 (three flavours, separate $\nu_e$ and $\nu_{\mu,\tau}$ temperatures, QED to $O(e^3)$). Left: the neutrino temperatures relative to the photon temperature; right: the comoving heating of the neutrinos by $e^+e^-$ annihilation.

## 8.5 The cosmic neutrino background

After decoupling the flavour coherence is lost: the CNB of UM-2 is an incoherent mixture of the three mass eigenstates with the Fermi-Dirac spectrum $f = 1/(e^{p/T_\nu}+1)$,

$$T_\nu = \left(\frac{4}{11}\right)^{1/3} T_{CMB} = 1.9454\ \text{K} = 1.676\times10^{-4}\ \text{eV}, \qquad n_i = \frac{3\zeta(3)}{2\pi^2}T_\nu^3 = 112.0\ \text{cm}^{-3}\ (\nu_i+\bar\nu_i),$$

$$\rho_i = 2\int \frac{d^3p}{(2\pi)^3}\sqrt{p^2+m_i^2}\,f, \qquad \Omega_\nu h^2 = \frac{\sum_i \rho_i}{\rho_c/h^2} = 6.382\times10^{-4} \quad (\rho_c/h^2 = 1.05368\times10^4\ \text{eV/cm}^3),$$

compared with the non-relativistic value $\sum m_i/94.06\ \text{eV} = 6.363\times10^{-4}$ at this temperature (the semi-relativistic $\nu_1$, whose energy density exceeds $m_1 n_1$ by 16 %, adds 0.3 %), with the conventional $\sum m_i/93.14\ \text{eV} = 6.426\times10^{-4}$ (which includes the heating of the non-instantaneous decoupling) and with $6.451\times10^{-4}$ for the heated spectrum of $N_{eff} = 3.0432$ (step 6). The states (DERIVED; mean velocity $\langle v\rangle = \int p^2(p/E)f\,dp/\int p^2 f\,dp$):

| state | mass (meV) | $n$ (cm$^{-3}$) | $\langle v\rangle/c$ today | $\lvert U_{ei}\rvert^2$ | $\lvert U_{\mu i}\rvert^2$ | $\lvert U_{\tau i}\rvert^2$ |
|---|---|---|---|---|---|---|
| $\nu_1$ | 1.000 | 112.0 | 0.435 | 0.677 | 0.112 | 0.211 |
| $\nu_2$ | 8.712 | 112.0 | 0.0604 | 0.301 | 0.428 | 0.271 |
| $\nu_3$ | 50.140 | 112.0 | 0.0105 | 0.022 | 0.460 | 0.518 |

The fold bands of dirac16complex are not part of the CNB (they are never thermalised, step 4).

## 8.6 Supernovae

**The MSW resonances of the envelope.** With $\rho(r) = 10^4\,(10^9\,\text{cm}/r)^3$ g/cm$^3$, $Y_e = 0.5$ (ASSUMED), a resonance lies at $n_e = \Delta m^2\cos2\theta/(2\sqrt{2}\,G_F E)$, i.e. at $r$ with $d\ln n_e/dr = -3/r$, its adiabaticity is $\gamma = \Delta m^2\sin^2 2\theta/(2E\cos2\theta\,|d\ln n_e/dr|)$ and the jump probability $P_{LZ} = e^{-\pi\gamma/2}$ (DERIVED):

| $E$ (MeV) | H (1-3): $\rho_{res}$ (g/cm$^3$) | $r_{res}$ (km) | $\gamma$ | L (1-2): $\rho_{res}$ (g/cm$^3$) | $r_{res}$ (km) | $\gamma$ |
|---|---|---|---|---|---|---|
| 10 | 3147 | $1.47\times10^4$ | 283 | 37.7 | $6.43\times10^4$ | 903 |
| 20 | 1573 | $1.85\times10^4$ | 178 | 18.8 | $8.10\times10^4$ | 569 |
| 40 | 787 | $2.33\times10^4$ | 112 | 9.42 | $1.02\times10^5$ | 358 |

Both resonances are adiabatic ($P_{LZ} \le 2.6\times10^{-77}$). For the normal ordering a $\nu_e$ produced in the core leaves the star as $\nu_3$, $P(\nu_e\to\nu_e) = |U_{e3}|^2 = 0.02215$, and an $\bar\nu_e$ as $\bar\nu_1$, $\bar P(\bar\nu_e\to\bar\nu_e) = |U_{e1}|^2 = 0.677$ (without collective effects and Earth matter). For representative accretion-phase neutrinospheres ($L = 6,\ 6,\ 4\times10^{52}$ erg/s and $\langle E\rangle = 11,\ 14,\ 15.5$ MeV for $\nu_e$, $\bar\nu_e$, $\nu_x$; ASSUMED) the mean energies observed at the Earth become 15.29 MeV ($\nu_e$) and 14.34 MeV ($\bar\nu_e$).

**The fast flavour instability.** With the ELN distribution $G(u) = \sqrt{2}G_F[n_{\nu_e}g_{\nu_e}(u) - n_{\bar\nu_e}g_{\bar\nu_e}(u)]$ in units of $\mu = \sqrt{2}G_F n_{\nu_e}$ ($G/\mu = g_{\nu_e} - \alpha g_{\bar\nu_e}$, $\alpha = n_{\bar\nu_e}/n_{\nu_e}$), the dispersion relation of the linearised equations, $\det\Pi = 0$ with $\Pi^{\mu\nu} = \eta^{\mu\nu} + \int du/2\,G\,v^\mu v^\nu/(\omega - ku)$, $v = (1,u)$, has the axisymmetric branch $(1+J_0)(-1+J_2) - J_1^2 = 0$, $J_n = \int du/2\,G\,u^n/(\omega-ku)$, and the azimuthal branch $-1 + \int du/2\,G(1-u^2)/(2(\omega-ku)) = 0$ ($\omega = \Omega - \Lambda^0$, $k = K - \Lambda^3$, $\Lambda^\mu = \int du/2\,G\,v^\mu$). The roots with $\text{Im}\,\omega > 0$ were found for real $k$ on a grid ($-1.2 \le k \le 1.2$, step 0.1) by vectorised Newton iterations with the analytic derivative from a grid of complex starting points (DERIVED):

| angular distributions | ELN crossing | largest growth (branch, $k$) | homogeneous mode | mode $K = \Lambda^3$ |
|---|---|---|---|---|
| $g_{\nu_e}\propto e^{-(1-u)^2/(2\cdot0.6^2)}$, $g_{\bar\nu_e}\propto e^{-(1-u)^2/(2\cdot0.4^2)}$, $\alpha = 0.8$ | yes | $0.0877\,\mu$ (axisymmetric, $k = -0.30$) | $0.0185\,\mu$ | stable |
| $g_{\nu_e} = 1+0.2u$, $g_{\bar\nu_e} = 1+0.6u$, $\alpha = 0.9$ | yes | $0.0979\,\mu$ (axisymmetric, $k = -0.10$) | stable | $0.0917\,\mu$ |
| as the first, $\alpha = 0.3$ | no | none | stable | stable |

(the largest azimuthal growth rates are $0.0200\,\mu$ and $0.0050\,\mu$ for the first two cases). Three independent evaluations agree to $1.2\times10^{-14}$: the analytic growth $\frac{1}{2}\sqrt{4I_1^2 - (I_0+I_2)^2}$ ($I_n = \int du/2\,G\,u^n$) of the mode $K = \Lambda^3$ against

the root of the dispersion relation at $k = 0$, and the eigenvalues of the discretised homogeneous equations $i\,dS_u/dt = (\Lambda^0 - u\Lambda^3)S_u - \int du'/2\,G(u')(1-uu')S_{u'}$ (300 nodes) against the root at $k = -\Lambda^3$. With an ELN crossing the gas is unstable, without one it is not, as Morinaga's theorem (2022) states; the second case shows that a stable homogeneous mode does not exclude growing inhomogeneous modes. Near the neutrinosphere, $n_{\nu_e} = 10^{32}$ to $10^{33}$ cm$^{-3}$ ($\mu = 1.27\times10^{-5}$ to $1.27\times10^{-4}$ eV), the first case grows by $e$ in $5.9\times10^{-10}$ to $5.9\times10^{-11}$ s over 17.8 to 1.78 cm: a million times faster than the hydrodynamics, which is why explosion models treat the conversion as instantaneous.

**Flavour conversion and the explosions.** The charged-current heating behind the shock is, schematically, proportional to $N\langle E^2\rangle$ of $\nu_e$ and $\bar\nu_e$ ($\langle E^2\rangle = \frac{5}{4}\langle E\rangle^2$ for spectra with the pinching $\alpha = 3$). For the neutrinospheres above, full three-flavour equipartition changes it by the factor 0.884, and pair conversion $\nu_e\bar\nu_e \to \nu_x\bar\nu_x$ that conserves the ELN until $\bar\nu_e$ is equipartitioned by 0.760 (DERIVED, schematic): the heavy-lepton neutrinos are more energetic but fewer and do not heat through charged currents. Gogilashvili and Tamborra (2026, arXiv:2605.16504) simulated 195 progenitors of 9 to 120 solar masses (GR1D, M1 transport of $\nu_e$, $\bar\nu_e$, $\nu_x$ in 18 energy bins, SFHo equation of state) with instantaneous flavour equipartition, conserving the electron lepton number and the number and momentum of the neutrinos in each energy bin, imposed at 0.02 s after bounce where the baryon density is below $\rho_c$ (STANDARD, cited):

| $\rho_c$ (g/cm$^3$) | no conversion | $10^9$ | $10^{10}$ | $10^{11}$ | $10^{13}$ |
|---|---|---|---|---|---|
| failed explosions | 25.6 % | 50.8 % | 77.4 % | 93.3 % | 96.4 % |
| weighted with the Salpeter mass function | 27.0 % | 47.6 % | 61.9 % | 73.9 % | 88.3 % |

The effect is strongest between 16 and 30 solar masses (easing the red-supergiant and supernova-rate problems), and the successful explosions leave lighter neutron stars (gravitational masses around 1.2 to 1.4 solar masses); the authors caution that the large fractions for deep conversion are in tension with observations and call for dynamical, multi-dimensional treatments.

# 9 Step 6: the CMB, the CNB and the BAO

The four-dimensional effective theory is general relativity (step 1) with photons, baryons, the three neutrino mass eigenstates, the dark matter (the odd quanta of $\Psi_A$ and $\Psi_B$) and $\Lambda_4$; there is no dark radiation. Its linear perturbations obey the standard Einstein-Boltzmann equations (Newtonian gauge, conformal time, $ds^2 = a^2[-(1+2\psi)d\tau^2 + (1-2\phi)d\vec{x}^{\,2}]$):

$$k^2\phi + 3\mathcal{H}(\phi' + \mathcal{H}\psi) = -4\pi G_N a^2\delta\rho, \qquad k^2(\phi' + \mathcal{H}\psi) = 4\pi G_N a^2(\bar\rho + \bar P)\theta,$$

$$\phi'' + \mathcal{H}(\psi' + 2\phi') + (2\mathcal{H}' + \mathcal{H}^2)\psi + \frac{k^2}{3}(\phi - \psi) = 4\pi G_N a^2\delta P, \qquad k^2(\phi - \psi) = 12\pi G_N a^2(\bar\rho + \bar P)\sigma,$$

with the cold dark matter $\delta_c' = -\theta_c + 3\phi'$, $\theta_c' = -\mathcal{H}\theta_c + k^2\psi$, the baryon-photon fluid with Thomson scattering, and one massive hierarchy per neutrino mass state $i$ (comoving momentum $q$, energy $E_i = \sqrt{q^2 + a^2m_i^2}$, multipoles $\mathcal{N}_l^{(i)}$ of the perturbed Fermi-Dirac distribution $\bar f_\nu$): $\mathcal{N}_0^{(i)\prime} = -\frac{qk}{E_i}\mathcal{N}_1^{(i)} - \phi'\frac{d\ln\bar f_\nu}{d\ln q}$, $\mathcal{N}_1^{(i)\prime} = \frac{qk}{3E_i}(\mathcal{N}_0^{(i)} - 2\mathcal{N}_2^{(i)}) - \frac{E_i k}{3q}\psi\frac{d\ln\bar f_\nu}{d\ln q}$, $\mathcal{N}_l^{(i)\prime} = \frac{qk}{(2l+1)E_i}[l\mathcal{N}_{l-1}^{(i)} - (l+1)\mathcal{N}_{l+1}^{(i)}]$ ($l \geq 2$). They were solved with CAMB 2.0.5 at the Planck 2018 values $\omega_b = 0.02237$, $\omega_c = 0.1200$, $100\theta_{MC} = 1.04092$, $\tau_{reio} = 0.0544$ (the optical depth to reionisation), $\ln(10^{10}A_s) = 3.044$, $n_s = 0.9649$ ($\theta$ held fixed, so $H_0$ is an output; $H_0$ in km/s/Mpc). In UM-2 the three mass states are evolved separately, each with the degeneracy $N_{eff}/3 = 1.0144$ (the heated spectrum); CAMB gives each the density $\deg\times\rho_{FD}(m';T_\nu)$ at $T_\nu = (4/11)^{1/3}T_{CMB}$, so its masses are $m_i' = m_i\deg^{-1/4}$, which reproduce the late-time density $m_i n_\nu \deg^{3/4}$ of the heated spectrum; CAMB finds $m_i'$ by inverting the full (relativistic) density from $\omega_\nu$ and the mass fractions, which were therefore iterated (13 iterations) until the masses agree to $7.8\times10^{-12}$ (DERIVED). The models are ΛCDM with the Planck 2018 parameters (one neutrino of 0.06 eV, $N_{eff} = 3.044$: the CAMB defaults), UM-2, and two variants with a thermalised fold band of dirac16complex (two Weyl species, which UM-2

does not contain; step 4), decoupled above the electroweak scale ($\Delta N_{eff} = 0.0958$) or just above the QCD crossover ($\Delta N_{eff} = 2(10.75/61.75)^{4/3} \times 1.0095 = 0.1963$) (DERIVED, `um2_cosmology.py`, 6 checks):

| model | $H_0$ | $\Omega_m$ | $N_{eff}$ | $Y_{He}$ | age (Gyr) | $z_*$ | $r_{drag}$ (Mpc) | BAO $\chi^2$ (13 points) |
|---|---|---|---|---|---|---|---|---|
| ΛCDM | 67.34 | 0.3154 | 3.044 | 0.24566 | 13.799 | 1089.91 | 147.103 | 32.55 |
| UM-2 | 67.37 | 0.3151 | 3.0432 | 0.24566 | 13.797 | 1089.91 | 147.109 | 31.72 |
| UM-2 + fold band (electroweak) | 68.48 | 0.3050 | 3.1390 | 0.24693 | 13.698 | 1090.00 | 146.630 | 15.43 |
| UM-2 + fold band (QCD) | 69.65 | 0.2948 | 3.2395 | 0.24824 | 13.596 | 1090.10 | 146.133 | 10.65 |

$\Omega_\nu h^2 = 6.4511 \times 10^{-4}$ in CAMB, equal to $6.4510 \times 10^{-4}$ from the CNB of step 5 times $(N_{eff}/3)^{3/4}$; the mass sum 0.0599 eV lies below the DESI DR2 bound 0.064 eV, which would allow $m_1 < 4.1$ meV. The UM-2 TT and EE spectra agree with Planck ΛCDM to $2.8 \times 10^{-4}$ ($30 \le \ell \le 2500$; for TT at most 0.007 of the cosmic variance $\sqrt{2/(2\ell+1)}$ of a single multipole; the binned full-sky cosmic-variance $\chi^2 = \sum_b (\bar\delta_b/\sigma_b)^2$ over bins $b$ of $\Delta\ell = 50$, with $\bar\delta_b$ the mean relative deviation and $\sigma_b = \sqrt{2/\sum_{\ell\in b}(2\ell+1)}$, is 0.023): UM-2 reproduces the CMB of ΛCDM. By themselves the three mass states instead of one state of 60 meV move TT and EE by $2.6 \times 10^{-4}$ and $2.3 \times 10^{-4}$, and $N_{eff} = 3.0432$ instead of 3.044 by $1.3 \times 10^{-4}$. The fold-band variants, at fixed Planck $\omega_b$, $\omega_c$, $\theta_*$, move TT by up to 1.5 % and 3.1 % (0.8 and 1.5 cosmic-variance units of a single multipole; binned $\chi^2 = 293$ and 1208): they are not fits to the CMB, and they lie 2.1 and 2.9 standard deviations above the ACT DR6 value $N_{eff} = 2.86 \pm 0.13$ (UM-2: 1.4). The CNB enters through $N_{eff} = 3.0432$ and the three neutrino masses.

## 10 Step 7: the dark-matter relic and its detection

The lightest odd states of $\Psi_A$ and of $\Psi_B$ are two degenerate Dirac fermions of mass $M_{DM} = 3.102204\, m = 44.35$ TeV ($m = H$), cold and neutral; UM-2 has no tip operator, so neither couples to the $Z$, and both are stable by the fold parity (step 4). Through I12 they annihilate in the p-wave into gauge-boson pairs: for $\mathcal{L} = G\,\bar\chi\chi F_{\mu\nu}F^{\mu\nu}$ per abelian channel, $\sigma v = G^2 s(s-4M^2)/(4\pi) \to G^2M^4v^2/\pi$, in total $\sigma v = (M^4v^2/\pi)(8G_g^2 + 3G_W^2 + G_B^2)$ (the polarisation sum $\sum|(k_1\cdot k_2)(e_1\cdot e_2) - (k_1\cdot e_2)(k_2\cdot e_1)|^2 = s^2/2$ and the spin trace $\mathrm{Tr}[(\not p_1 + M)(\not p_2 - M)] = 2(s-4M^2)$ checked numerically). Since $\bar\Psi\Psi = \bar\Psi_A\Psi_A + \bar\Psi_B\Psi_B$ does not mix the halves, each species freezes out separately and contributes half of $\Omega_c h^2 = 0.120$. Freeze-out (Dirac fermion, $g_* = 105.75$ at $T_f = M/x_f \approx 1.5$ TeV: the Standard Model of three generations without a Higgs particle; the right-handed neutrinos and the fold bands are never thermalised):

$$x_f = \ln\left[0.038\, c(c+2)\, g\, M_{Pl} M \langle\sigma v\rangle / \sqrt{g_* x_f}\right] = 30.38, \qquad \Omega h^2 = \frac{2 \times 1.07 \times 10^9\ \mathrm{GeV}^{-1}\, x_f}{\sqrt{g_*}\, M_{Pl}\, 3b/x_f} = 0.060 \text{ per species}$$

($\langle\sigma v\rangle = 6b/x$, $c = 1/2$, $g = 2$, $M_{Pl} = G_N^{-1/2} = 1.22091 \times 10^{19}$ GeV, standard radiation-dominated freeze-out) requires $\langle\sigma v\rangle(x_f) = 2.01 \times 10^{-25}$ cm$^3$/s per species, 87 % of the s-wave thermal unitarity estimate $(4\pi/M^2)\sqrt{x_f/\pi} = 2.32 \times 10^{-25}$ cm$^3$/s (an estimate made for an s-wave process, applied here to a p-wave one), i.e.

$$(8G_g^2 + 3G_W^2 + G_B^2)^{1/2} = (15.5\ \mathrm{TeV})^{-3}, \qquad G_i = \mathcal{O}\, c_i g_i^2/\Lambda_\star^3, \quad \mathcal{O} = 0.7072$$

(a single species carrying all the dark matter, as in version 1 where the odd $\Psi_A$ states decayed, would need $(17.6\ \mathrm{TeV})^{-3}$). The portal scale lies below the dark-matter mass: the effective description of I12 is at its edge (an ultraviolet completion is OPEN). Direct detection (ESTIMATE): even if the whole annihilation went into gluons, the trace anomaly ($\langle N|G^\alpha G^\alpha|N\rangle = -\frac{8\pi}{9\alpha_s} m_N f_{TG}$) gives $\sigma_{SI} = \mu^2 f_N^2/\pi = 6.3 \times 10^{-53}$ cm$^2$ for each half (the two halves have the same cross-section and together the full local density, so the limit applies as for one species), eight orders below the LZ limit extrapolated to 44 TeV ($2.4 \times 10^{-45}$ cm$^2$); the electroweak channels have no tree-level

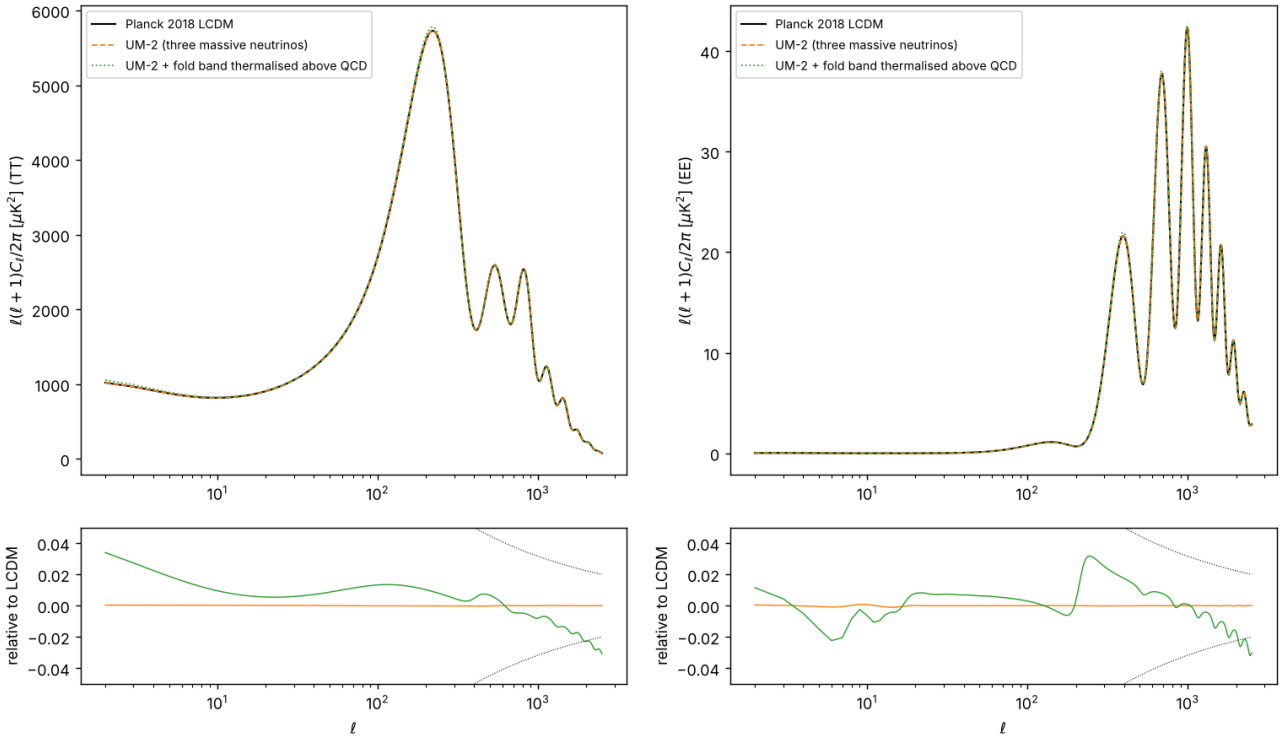



Figure 4: CMB temperature (left) and E-mode (right) spectra of UM-2 (three massive neutrinos), of Planck 2018 ΛCDM and of the variant with a fold band thermalised above the QCD crossover; lower panels: relative difference to ΛCDM with the cosmic variance of one multipole (dotted).

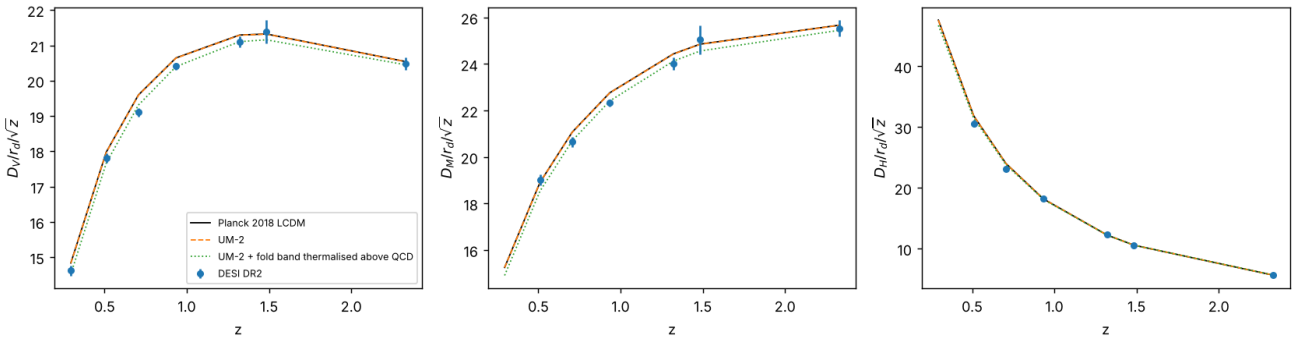



Figure 5: BAO distances divided by $\sqrt{z}$ against the DESI DR2 measurements; the $\chi^2$ uses $D_V$ for BGS and $D_M$, $D_H$ with their correlation for the other tracers (the $D_V$ points of the anisotropic tracers are shown for illustration).

nucleon coupling. The relic value is conditional: if the radion oscillation left over from the deflation (which redshifts like dust) dominated the Universe until it decayed, it would dilute the dark matter after freeze-out (the moduli problem, below).

# 11 Accuracy of the solution

- Exact (symbolic): the background equations, the static block, $G(k_*)$, the radion frequency, the first-order block response, the dimensional reduction, the field reductions and the zero modes.
- Numerical, with measured errors: boson levels (mpmath, 30 digits; finite elements $\leq 5 \times 10^{-10}$); fermion masses (exact roots; the first-order formula within $1.1 \times 10^{-3}$ for the top quark and $3 \times 10^{-4}$ for every other fermion; shooting $\leq 6.1 \times 10^{-10}$); the three-generation tip determinant (mpmath, 50 digits; agreement with the one-generation roots $3.0 \times 10^{-42}$); Kohn-Sham energies (Richardson errors below $1.5 \times 10^{-9}$ relative; conservation $9 \times 10^{-8}$); primordial integrations (constraint below $6 \times 10^{-10}$); flavour evolution (vacuum formula $3.9 \times 10^{-15}$; solar layers converged to $2.2 \times 10^{-6}$; the 1-3 resonance formula $1.6 \times 10^{-4}$); neutrino decoupling (LSODA, relative tolerance $10^{-11}$; the method reproduces the published values to $10^{-3}$); fast flavour growth rates ($1.2 \times 10^{-14}$ between three evaluations); CMB (CAMB default accuracy, lensing included; the three neutrino masses to $7.8 \times 10^{-12}$).
- Approximations: the static block with an assumed stabiliser; the good sector (no extra-time momenta); the Kohn-Sham mean-field treatment (local exchange of the contact interaction); portal couplings as external mean fields; tree level for the electroweak widths; the coherent (collisionless) flavour evolution in the Sun, the Earth and the supernova envelope, with schematic density profiles; neutrino decoupling with one temperature per flavour (momentum averaged); the linear analysis of the fast flavour instability and a schematic estimate of its effect on the heating; the freeze-out formula; order $\epsilon$ for the cosmology.

# 12 What the solution does not reproduce, and what is open

- **Conflicts with data, inherent in the order**: no Higgs boson (one is observed at 125 GeV), and without it the longitudinal $W^+W^-$ scattering violates perturbative unitarity near $\sqrt{8\pi}\, v \approx 1.2$ TeV, far below the Kaluza-Klein modes. (The one-generation conflict of version 1, a total $Z$ width of 1.23 GeV, is removed: three generations give 2.485 GeV against the measured 2.4955 GeV.)
- **Open**: the origin of the stabiliser; the extra-time modes outside the good sector (tachyonic for momentum $q > fM$); the closed time-like curves implied by compact extra times; the radion left over from the deflation (ESTIMATE: a Planck-suppressed decay $\Gamma = C_r\, m_r^3/(4\pi \bar{M}_{Pl}^2)$, with the reduced Planck mass $\bar{M}_{Pl} = 2.435 \times 10^{18}$ GeV and an order-one coupling factor $C_r$, gives the lifetime $0.43\,\mathrm{s}/C_r$ and the reheating temperature $T_{RH} = (90/(\pi^2 g_*))^{1/4}\sqrt{\Gamma \bar{M}_{Pl}} = 1.85\sqrt{C_r}$ MeV, $g_* = 10.75$, below the 4.7 MeV that nucleosynthesis needs unless $C_r > 6.5$: a moduli problem); the block-shape mode driven by $p_0 - p_t$; the tuning of $\Lambda_4$ ($10^{-92} H^2$); the Kaluza-Klein graviton spectrum with the Lovelock corrections; the electroweak $S$ parameter of the Higgsless tip and the electroweak loop corrections; anomalies of the eight-dimensional bulk theory; the absolute neutrino mass scale and the mass ordering (bounded and preferred by data, chosen here); the collective flavour evolution of supernova neutrinos beyond the linear analysis; the ultraviolet completion of I12 (the relic needs a cross-section close to the unitarity estimate); the joint CMB and BAO fit of the fold-band variants.